17 April 2026 09:00

Built-in Modules

OS module -->it provides a way to interact with the operating system using python code . It allows us to work with:-

1. **Files and folders**
2. **Paths**
3. **Environmental variables**
4. **Processes**
5. **Permissions**
6. **System commands**
7. **Directory navigation**

Import os

The os module gives python programs access to OS- dependent functionalities

Its communicate with
Windows/
Linux
Macos

Example:

```
print(os.name) # windows-nt    mac/linux ---posix it gives the name of operating system
```

Wy use os module?

Use case	Example
File management	Create/delete folders
Automation	Rename /create 1000+ files
Deployment	.env reading the environmental variable
Security	Manage the permission
Data science	Navigate project folders
DevOps	Run shell commands
Backed Apps	Access system configuration

Directory operations:
os.getcwd() --> return current working directory

os.chdir(path) --> change current directory

```
Os.listdir()--> used to list files/folders

os.mkdir("reports")--> making directory -- used to
create folder
os.makedirs("2026/April/Reports") #create nested
folder
 os.rmdir("Day10") #--> remove empty folder

os.removedirs("2026/April/Reports") #to remove
nested directories
```

File operations :-

```
# os.rename("oldfilename.extention","new filename.extention")
# os.rename("wipro_tra.txt","wipro_trainings.txt")

# os.remove("filename.extention")
os.remove("wipro_trainings.txt")

print(os.path.exists("day21.py"))  #check if the file is exists or not , it returns true if file is exists  else false
```

Path handlings:-

Os.path.exists("/etc/nginx/nginx.cong")

```
# print(os.path.exists("D:\Myworkspace\day21.py"))
# path=os.path.join("Users","Admin","Docs") #create safely join paths
# print(path)
# print(os.path.isfile("day1.py")) # check file is available or not
print(os.path.isdir("D:\Myworkspace\DAY8")) # CHECK THE DIRECTORY IS AVAILABLE OR NOT IN THE PATH
```

environmental variables :-

Used to store sensitive or system settings :-

Examples;
API keys
Database password
Usernames
Path

```
#pip install python-dotenv
# os.getenv()
from dotenv import load_dotenv
load_dotenv() # load .env file
# dbserv=os.getenv("DB_SERVER_USER")
# dbhost=os.getenv("DB_SERVER_HOST")
# dbpassword=os.getenv("DB_SERVER_PASSWORD")
# print(f"user : {dbserv} hostname:{dbhost} passwordv : {dbpassword}")
#os.environ --> set environmet varianle named ENV with value for the current python process
# os.environ["ENV"] = "production"
# print(os.getenv("ENV"))
# var=os.walk(os.getcwd()) # used to get the information of each and every folders and files as nested format
# print(var)
# for  value in var:
#     print(value)
# var=os.walk(os.getcwd())
# for cur_dir,list_dir,list_files in var:
#     print(cur_dir)
#     print(list_dir)
#     print(list_files)
```

Sys module:-
The sys is a system-specific parameters and functions module, it helps your python program communicate with the interpreter and operating system environment

Used for:
1. **reading the command line arguments**

2. Exiting the script safely
3. Redirecting the output
4. Accessing the python path
5. Error logging
6. Automate the scripts

Command line arguments (we are passing arguments at the time of program compilation)

Python filename.py -->

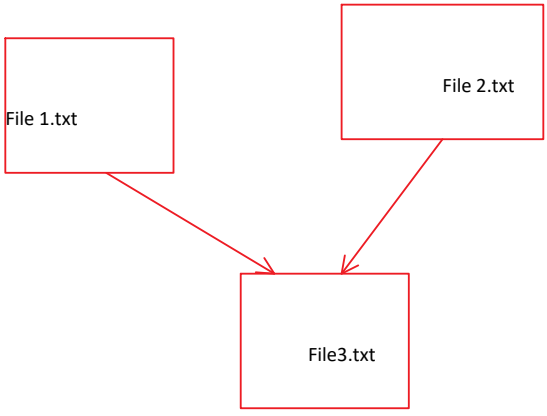
Python filename.py 10 20 30 40 these are called as commandline args

By using sys module we can read command line arguments
Argv --> variable , list type (ordered list and allow duplicate values)

	0	1	2	3
Python	day3.py	10	20	30

0th index always consider as file name

When cmd line arguments reading in to python program initially the type is string type only .




```
from  sys import argv
# print(type(sys.argv))
# print(sys.argv[1])
# print(sys.argv[2])
# print(f"the number of cmd line arguments {len(argv)}")
# print(f"list of cmd arguments {argv}")
# print(f"the cmd arguments one by one :")
# for arg in argv:
#     print(arg)
# print sum of all cmd line arguments:
# args=argv[1:]
# sum=0
# for x in args:
#     sum=sum + int(x)
# print(sum)
```

```
Case1 ;-
The space between two values is considered

Murali mohan
```

PS D:\Myworkspace> python .\day3.py 'murali mohan'

Case 2:

```
# print(argv[1] + argv[2])
print(int(argv[1]) + int(argv[2]))
```

Case 3:
PS D:\Myworkspace> python .\day3.py 20 30
Traceback (most recent call last):
  File "D:\Myworkspace\day3.py", line 113, in <module>
    print(argv[100])
        ~~~~^^^^^^
        ~~~~

IndexError: list index out of range

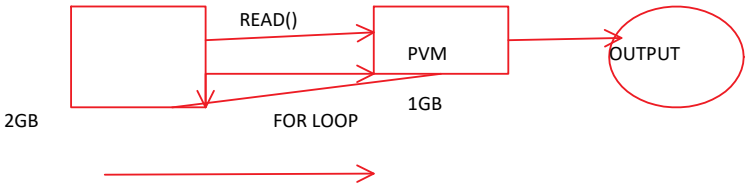
```
print(argv[100])

env=argv[1]
if env=="prod":
    print("deployong to the production")
elif env == "dev":
    print("deploying  to development")
```

File operations:-

File IO means reading data from files and writing data to files .
Python having built-in function **Open()** give access to any file on disk
With statement ensures the file is properly closed after use -- even if an error occurs .

MODE	DESCRIPTION
'r'	**READ ONLY(default). File must exists**
'w'	**Write . Create file if does not exist . Overwrite the file if exists**
'a'	**Append . Adds to end of file without deleting the content**
'r+'	**Read and write . File must exists**
'x'	**Exclusive creation . Fails if fail already exists**



Read operations

```
# f=open(r'C:\Users\babud\Desktop\wipro-april\questionpaper.txt','r')
# print(f.read())
# f.close()
# using context manager  to open file
# with open(r'C:\Users\babud\Desktop\wipro-april\questionpaper.txt','r') as
f:
#     for line in f:
#         print(line,end='')
# with open(r'C:\Users\babud\Desktop\wipro-april\questionpaper.txt','r') as
f:
#     print(f.read()) read() can  read the file data at once , if 1Gb of
file uhave u need memory space gb
# with open(r'C:\Users\babud\Desktop\wipro-april\questionpaper.txt','r') as
f:
#     # print(f.read(20)) # print top 20 characters
#     print(f.readlines()) # print each line consists of one list

# write operation
# with open('TEST.txt','w') as f:
#     f.write(" hello ,")
# with open('TEST.txt','a') as f:
#     f.write(" \n i am murali mohan M")
with open(r'C:\Users\babud\Desktop\wipro-april\questionpaper.txt','r') as
readfile:
    for line in readfile:
        if line.startswith("Answer") is True:
            with open('Answers.txt','a') as appendfile:
                appendfile.write(line)
```

```
import math
import random
import sys

sys.path.append(r'c:/Users/babud/Desktop/wipro3/day4')

import function_python

print(function_python.fact(6))  # Testing import
print(math.sqrt(10))
print(math.pow(2, 3))
```

```
print(math.ceil(4.9))
```